Molly Wolfson

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Overview

I am a CCAPP Postdoctoral Fellow and President's Postdoctoral Scholar at The Ohio State University. My research focuses on cosmology and the high-redshift universe, using the Lyman-α forest to study the intergalactic medium (IGM), the epoch of reionization, and the thermal and ionization history of the Universe. I have authored seven first-author publications (five published, one submitted, and one under collaboration review), three publications as a second or third author with significant contributions, and contributed to 12 collaboration publications, including nine from DESI, two from XQR-30, and one from GHOSTLy.

Education

University of California, Santa Barbara — Santa Barbara, CA
- Physics M.A. 2021; Physics Ph.D. 2024

The University of Chicago — Chicago, IL
- Physics B.A. with honors; Mathematics B.S. 2018

Employment

2025–present: **President's Postdoctoral Scholar**, OSU, Columbus, OH

2024–present: **CCAPP Postdoctoral Fellow**, OSU, Columbus, OH

First-Author Publications

1. **Wolfson, M.**, et al. "Measurements of Lyman-α forest 1D power spectrum at $5 \leq z \leq 6$ from XQR-30" 2026, under XQR-30 collaboration review
2. **Wolfson, M.**, Hennawi, J. F., Davies, F. B., Lukić, Z. "A Robust Estimator for the Lyman-α Forest 1D Power Spectrum at High Redshift" 2026, arxiv:2610.31626, submitted to J. Cosmology Astropart. Phys.
3. **Wolfson, M.**, Hennawi, J. F., Davies, F. B., Lukić, Z., Oñorbe, J. "Forecasting constraints on the high-z IGM thermal state from the Lyman-α forest flux auto-correlation function" 2025, MNRAS, 540 (2), 1412-1431
4. **Wolfson, M.**, et al. "Measurements of the $z > 5$ Lyman-α forest flux auto-correlation functions from the extended XQR-30 data set" 2024, MNRAS, 531 (3), 3069-3087
5. **Wolfson, M.**, Hennawi, J. F., Davies, F. B., Oñorbe, J. "Forecasting constraints on the mean free path of ionizing photons at $z \geq 5.4$ from the Lyman-α forest flux auto-correlation function" 2023, MNRAS, 521 (3), 4056-4073
6. **Wolfson, M.**, Hennawi, J. F., Davies, F. B., Oñorbe, J., Hiss, H., Lukić, Z. "Improving IGM temperature constraints using wavelet analysis on high-redshift quasars" 2021, MNRAS, 508 (4), 5493-5513
7. **Wolfson, M.**, Liepold, E. R., Lin, B., and Rice, S. A. "A comment on the position dependent diffusion coefficient representation of structural heterogeneity" 2018, J. Chem. Phys., 148 (19), 194901

Contributing-Author Publications

1. González-Hernández, D., Doughty, C., **Wolfson, M.**, Hennawi, J. F., Jin, Z. "Using Neural Emulators and Hamiltonian Monte Carlo to constrain the Epoch of Reionization's History with the Lyα Forest Power Spectrum" 2025, MNRAS, 550 (2), stag1233
2. Brodzeller, A., **Wolfson, M.**, et al. "Construction of the damped Lyα absorber catalog for DESI DR2 Lyα BAO" 2025, Phys. Rev. D, 112 (8), 083510
3. Jin, Z., **Wolfson, M.**, Hennawi, J. F., González-Hernández, D., "Neural network emulator to constrain the high-z IGM thermal state from Lyman-α forest flux auto-correlation function" 2025, MNRAS, 536 (3), 2277-2293

Collaboration Papers

1. Bosman, S. E. I. et al. including **Wolfson, M.** "Nearly pristine gas in a galaxy 1 billion years after the Big Bang" 2026, arxiv:2610.16509, accepted in Nature
2. Herbold, M. et al. including **Wolfson, M.** "Validation of the DESI DR2 Lyα forest full-shape analysis" 2026, arxiv:2607.27411
3. DESI Collaboration including **Wolfson, M.**, "DESI DR2 Results IV: Alcock-Paczyński Measurements from the Lyman Alpha Forest and Cosmological Constraints" 2026, arxiv:2607.27410
4. Cases, L. et al. including **Wolfson, M.** "Validation of the DESI DR2 Lyα BAO analysis using synthetic datasets" 2026, Phys. Rev. D, 113 (12), 023520
5. DESI Collaboration including **Wolfson, M.**, "DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints" 2025, Phys. Rev. D, 112 (8), 083515
6. DESI Collaboration including **Wolfson, M.**, "DESI DR2 results. I. Baryon acoustic oscillations from the Lyman alpha forest" 2025, Phys. Rev. D, 112 (8), 083514
7. Elbers, W. et al. including **Wolfson, M.** "Constraints on Neutrino Physics from DESI DR2 BAO and DR1 Full Shape" 2025, Phys. Rev. D, 112(8), 083513
8. Andrade, U. et al. including **Wolfson, M.** "Validation of the DESI DR2 Measurements of Baryon Acoustic Oscillations from Galaxies and Quasars" 2025, Phys. Rev. D, 112 (8), 083512
9. Lodha, K. et al. including **Wolfson, M.** "Extended Dark Energy analysis using DESI DR2 BAO measurements" 2025, Phys. Rev. D, 112 (8), 083511
10. DESI Collaboration including **Wolfson, M.**, "Data Release 1 of the Dark Energy Spectroscopic Instrument" 2026, AJ, 171 (5), 285
11. D'Odorico, V., et al. including **Wolfson, M.**, "XQR-30: the ultimate XSHOOTER quasar sample at the reionization epoch" 2023, MNRAS, 523 (1), 1399-1420
12. Zhu, Y. et al. including **Wolfson, M.**, Yang, J., "Damping Wing-Like Features in the Stacked Lyα Forest: Potential Neutral Hydrogen Islands at $z < 6$" 2024, MNRAS, 533 (1), L49-L56

Conference Proceedings

1. González-Hernández, D., **Wolfson, M.**, Hennawi, J. F., "Reliable Parameter Inference for the Epoch of Reionization using Balanced Neural Ratio Estimation" Machine Learning and the Physical Sciences Workshop at NeurIPS, December 6th, 2025, San Diego, CA, arXiv:2511.02808
2. González-Hernández, D., **Wolfson, M.**, Hennawi, J. F., "Accelerating statistical inferences in astrophysics with Neural Networks and Hamiltonian Monte Carlo" 2024 ICML Workshop on Structured

Probabilistic Inference & Generative Modeling, Vienna, Austria, https://openreview.net/forum?id=uKxqQ4DoFE

Awards and Honors

OSU President's Postdoctoral Scholar	2025–2027
Worster Summer Research Fellowship	2023
Doctoral Student Travel Grant	2023
Mananya Tantiwiwat Fellowship Award	2022
UCSB Department of Physics, Department Service Award	2018–2020
University of Chicago Dean's List	2014–2018
Enrico Fermi Institute Undergraduate Summer Research Grant	2017
UCISTEM Summer Research Grant	2015

Research Talks

Invited

2027 *Upcoming* – Lyman-α Forest Cosmology Workshop, January 2027, Riverside, CA

2026 Institute for Theory and Computation Luncheon, October 2026, Cambridge, MA

2026 Galaxies Seminar at the University of Michigan, March 2026, Ann Arbor, MI

2024 Center for Cosmology and AstroParticle Physics Seminar, June 2024, Columbus, OH

2023 Berkeley Center for Cosmological Physics Seminar, October 2023, Berkeley, CA

2022 Berkeley Center for Cosmological Physics Seminar, October 2022, Berkeley, CA

Contributed

2026 CCAPP Symposium, September 2026, Columbus, OH

2026 Ohio Cosmic Connections, May 2026, Cincinnati, OH

2026 *Towards the Full Bloom of First Star, Galaxy, and Black Hole Formation*, March 2026, Tokyo, Japan

2025 DESI Collaboration Meeting, December 2025, Tucson, AZ, USA

2025 *Cosmo-25*, October 2025, Pittsburgh, PA

2025 Institut de Física d'Altes Energies Seminar, September 2025, Barcelona, Spain

2025 *Lyman-α forest full shape analysis with DESI Year 3*, September 2025, Columbus, OH

2025 CCAPP Symposium, September 2025, Columbus OH

2024 DESI Collaboration Meeting, December 2024, Cancun, Mexico

2024 Lyman-α Workshop, October 2024, Columbus, OH

2024 CCAPP Symposium, September 2024, Columbus, OH

2024 *243th Meeting of the American Astronomical Society*, January 2024, New Orleans, LA

2023 MIT Monday Afternoon Talk, July 2023, Cambridge, MA

2023 *Reionization in the Summer*, June 2023, Heidelberg, Germany

2023 UCSB Astro Lunch Seminar, May 2023, Santa Barbara, CA

2023 *Future Cosmology*, April 2023, IESC Cargese, France

2022 *240th Meeting of the American Astronomical Society*, June 2022, Pasadena, CA

2022 *Reionization and Cosmic Dawn: Looking Forward to the Past*, March 2022, Berkeley, CA

Student Supervision

Summer 2025: **Summer Undergraduate Research Program (SURP) Mentor** Ohio State University
Supervised Brandon Lam (undergrad)

2022–present: **Research Mentor** University of California, Santa Barbara
Supervised Diego González-Hernández (continuing PhD student)
Resulted in publication González-Hernández et al. (2025) and two conference proceedings

2022–2024: **Research Mentor** University of California, Santa Barbara
Supervised Linda (Zhenyu) Jin (undergrad, now PhD student at University of Washington)
Resulted in publication Jin et al. (2025)

Teaching Experience

2018–2019: **Teaching Assistant** UCSB Department of Physics
PHYS 3L (now 20AL), PHYS 4L (now 20BL) - introductory labs for physics majors

2016–2018: **Physics Core Tutor** The University of Chicago Harper Library
Covered introductory physics course material and beyond

Synergistic Activities

1. **Professional Service:**
   (i) Referee, A&Aand MNRAS
   (ii) Organizer, "Cosmo Lunch", journal club, OSU, Columbus, OH, 2025 – 2026
   (iii) SOC Member, "Lyman-α forest full shape analysis with DESI Year 3", OSU, Columbus, OH, 2025
   (iv) Organizer, "CCAPP Seminar" talk series, OSU, Columbus, OH, 2024 – 2025
   (v) SOC Member, "Lyman-α Workshop" OSU, Columbus, OH, 2024
   (vi) Organizer, "Astro Lunch" talk series, UCSB, Santa Barbara, CA 2019 – 2024
2. **Collaborations:**
   (i) Member, the DESI collaboration, 2024 – present
   (ii) Co-Investigator, GHOSTLy, 2024 – present `https://www.gemini.edu/observing/phase-i-proposing-time/llp/approved-llps/ghostly-quasar-sample`
   (iii) Member, the XQR-30 team, 2021 – present `https://xqr30.inaf.it/`
3. **Public Talks:**
   (i) "Exploring the Epoch of Reionization through High-Redshift Quasars" FOSAA, October 2025
   (ii) "Cosmology 101" for OSU ASPIRE, June 2025
   (iii) "Cosmic Lighthouses: Navigating the Early Universe with Quasars", UCSB Grad Slam, March 2023
   (iv) "The History of the Universe with High-Redshift Quasars", UCSB Lunch & Learn, June 2023
4. **UCSB Mentoring and Leadership:**
   (i) Mentor, Graduate Scholars Program, 2020 – 2024
   (ii) Member, Women and Gender Minorities in Physics, 2018 – 2024
   (iii) President, Women and Gender Minorities in Physics, 2019 - 2022
   (iv) Mentoring Chair, UCSB Physics GradLife, 2019 - 2021
   (vi) Mentor, Women in Science and Engineer Mentoring Program, 2018 – 2019
5. **Invited Panels:**
   (i) OSU Momentum Physics Pathways Panel, September 2026
   (ii) OSU ASPIRE Graduate School Panel, June 2025
   (iii) "Being a Woman in Physics" UCSB SPS, 2021
   (iv) "Applying to Graduate School and Fellowships" APS CUWiP, January 2019
   (v) "Exploring Undergraduate Research Opportunities" UCSB Dept. of Physics, 2018